Lab D: Exp 24. Identifying the Structure of an Aldehyde by Quantitative Analysis

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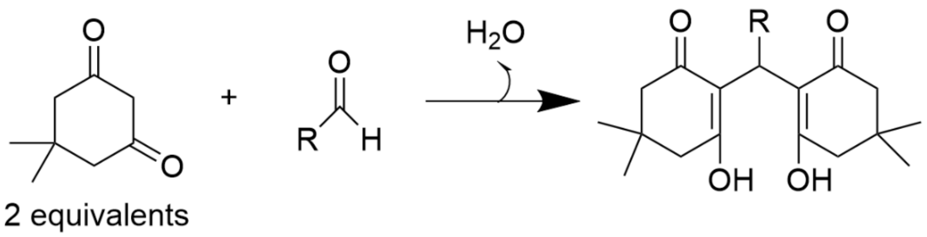
I. INTRODUCTION

Aldehydes are an important group of organic compounds that undergo a wide variety of chemical reactions.[1] Aldehydes are also used in the production of perfumes, plastics, dyes, and pharmaceuticals.[2] The reaction between dimedone and various benzaldehyde derivatives to form dimethone anhydride derivatives is very significant because the products have been shown to exhibit biological applications as they hold antibacterial, antiviral, and anticancer properties.[3] Methods of identifying an unknown aldehyde and its structure include ^{1}H NMR and ^{13}C NMR analysis, gas chromatography followed by mass spectrometry, or alternatively liquid chromatography followed by mass spectrometry. However, these methods are expensive and require advanced equipment, thus they are less practical. Therefore, the chosen method for unknown aldehyde identification used in this investigation is melting point analysis because of its high accuracy, reliability, and low cost.

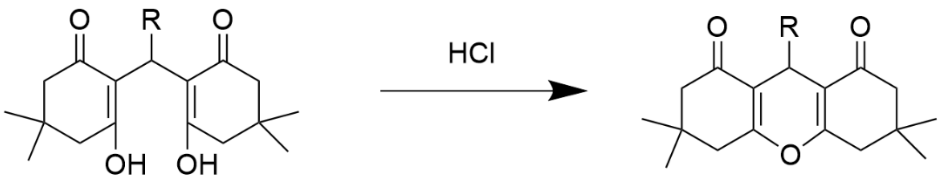
An aldehyde was reacted with an excess of dimedone to form the dimethone derivative, the general scheme of this reaction is outlined in *Scheme 1* below.

Scheme 1: General overview of the synthesis of the dimethone derivative



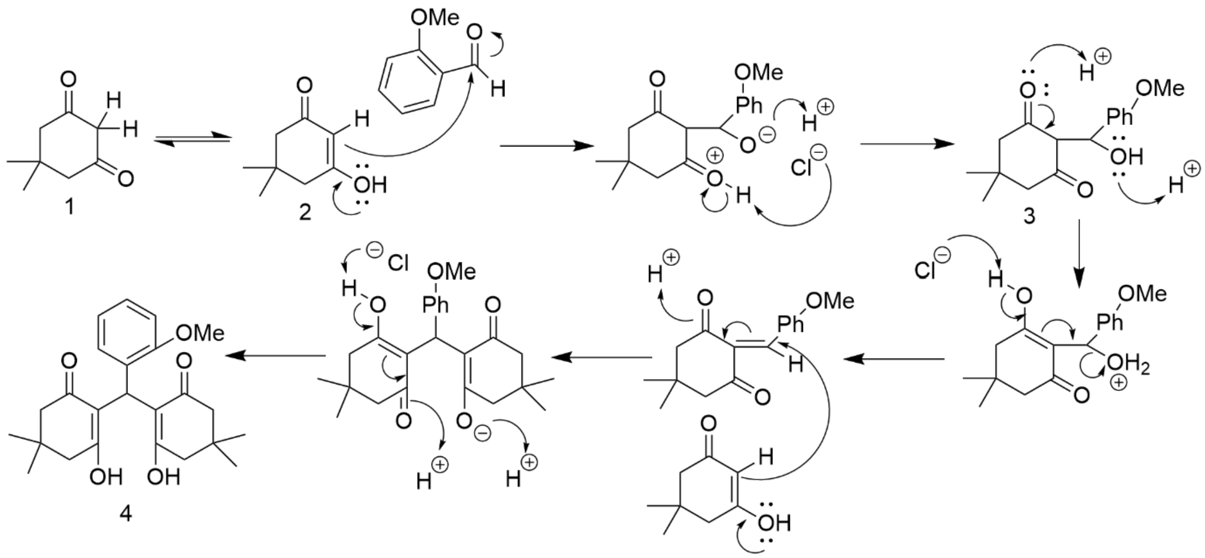
The dimethone derivative was then converted to the dimethone anhydride derivative using an acid catalyst, the general scheme of this reaction is outlines in *Scheme 2* below.

Scheme 2: General overview of the dehydration of the dimethone derivative to produce the dimethone anhydride derivative



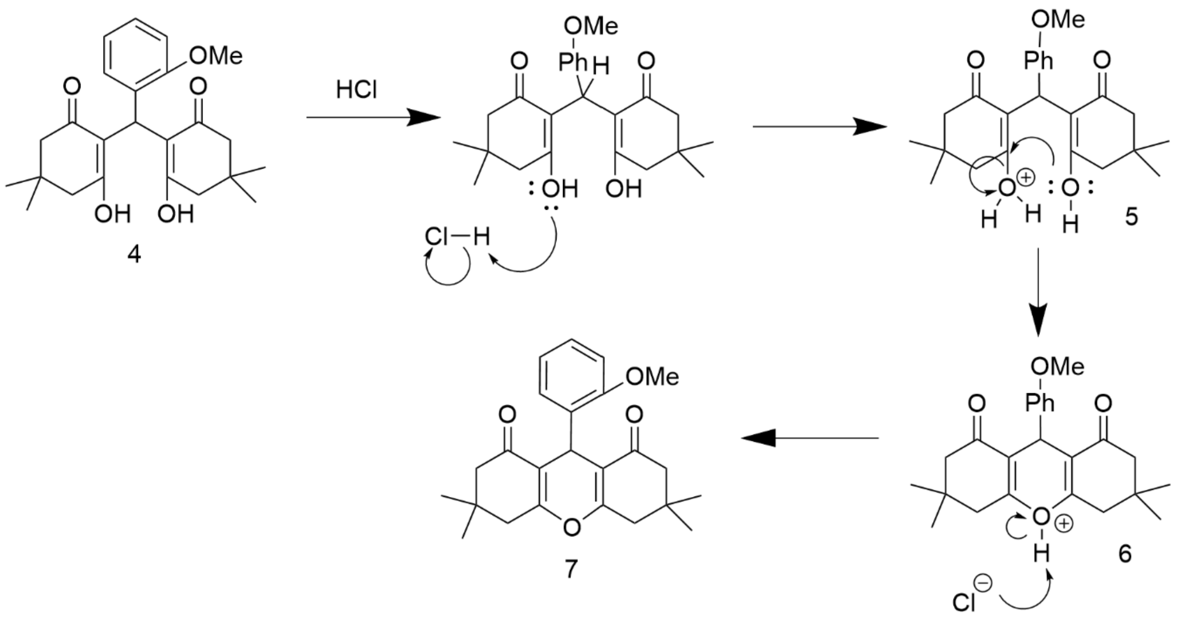
Based on melting point analysis, the unknown aldehyde #7 used in this investigation was identified to be 2-methoxybenzaldehyde. The specific reaction mechanism conducted in this investigation is outlined in *Scheme 3* below.

Scheme 3: Mechanism of 2-methoxybenzaldehyde and dimedone to form dimethone derivative



The product of reacting 2-methoxybenzaldehyde with excess dimedone (1) is 2,2'-((2-methoxyphenyl) methylene)bis(3-hydroxy-5,5-dimethylcyclohex-2-en-1-one) (4). Initially, dimedone underwent tautomerisation from the keto form (1) to the enol form (2). The dimedone in enol form attacked the carbonyl carbon on the 2-methoxybenzaldehyde to produce an intermediate. Another dimedone in enol form attacked the aldehyde carbon on the intermediate to produce the enol form of the product. The enol form then underwent tautomerisation to form the keto product (4). The investigation continued as product (4) was dehydrated using hydrochloric acid catalyst to form product (7), the mechanism for this reaction is outlined in *Scheme 4* below.

Scheme 4: Mechanism of the dehydration of the dimethone derivative product (4) to produce the dimethone anhydride derivative product (6)



Product (4) is pronated by the hydrochloric acid catalyst to form the positively charged intermediate (5). The neutral hydroxyl group attacks the opposite alpha carbon forming intermediate (6) and eliminating water from the molecule. The chlorine ion deprotonates the positively charged oxygen and this forms the anhydrous product, 9-(2-methoxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene -1,8(2H)-dione (6).

II. EXPERIMENTAL PROCEDURE

This experiment was conducted by Mark Rejna and Ali Mozaffar following instructions outlined in the *Understanding the Principles of Organic Chemistry* textbook.[4]

5,5-dimethyl-1,3-cyclohexanedione (dimedone) (1)

^{1}H NMR (400 MHz, $CDCl_3$) δ 1.1 (s, 12H), 2.2 (s, 4H), 2.5 (s, 4H), 3.4 (s 1H), 5.5 (s 1H), 9.8 (s, 1H)

^{1}H NMR (400 MHz, $(CD_3)_2SO$) δ 1 (s, 6H), 2.1 (s, 4H), 5.3 (s, 1H), 11 (s, 1H)

2-methoxybenzaldehyde

^{1}H NMR (400 MHz, CDCl3) δ ~4 (s, 3H), 6-8 (d, J = 6-10 Hz, 1H), 6-8 (dd, J = 6-10 Hz, 1H), 6-8 (dd, J = 6-10 Hz, 1H), 6-8 (d, J = 6-10 Hz, 1H), 10 (s, 1H)

2,2'-((2-methoxyphenyl)methylene)bis(3-hydroxy-5,5-dimethylcyclohex-2-en-1-one) (4)

2-methoxybenzaldehyde (100.2 mg, 0.736 mmol), dimedone (300.1 mg, 2.14 mmol), and a 1:1 ethanol:water mixture (4ml) were combined into a large test tube. The melting range of the reactant 2-methoxybenzaldehyde was measured to be 38 to 40 °C. The reaction mixture was heated at reflux for 5 minutes, the mixture turned a clear yellow color. The reaction mixture was then left to cool to room temperature. White precipitate/crystals were present at the bottom of the test tube with the yellow solution above. The reaction mixture was then cooled in an ice-water bath for 10 minutes. The resulting crystals that formed after cooling were isolated on a Hirsch funnel, washed with an ice cold 1:1 ethanol:water mixture, and dried by drawing a vacuum through the filtrate. The yield of the resulting crystal product, 2,2'-((2-methoxyphenyl)methylene)bis(3-hydroxy-5,5-dimethylcyclohex-2-en-1-one), was measured (372.9 mg, 257.54 %). The melting range of the product was measured to be between 187 and 189.4 °C.

[1]H NMR (400 MHz, $CDCl_3$) δ 0.7-1.3 (s, 3H), 1.2-1.4 (s, 4H), 1.2-1.4 (s, 4H), 1.4-1.7 (s, 1H), 2 (s, 12H), 1-5 (s, 2H), 6-8 (d, J = 6-10 Hz, 1H), 6-8 (dd, J = 6-10 Hz, 1H), 6-8 (dd, J = 6-10 Hz, 1H), 6-8 (d, J = 6-10 Hz, 1H)

9-(2-methoxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene -1,8(2H)-dione (6)

2,2'-((2-methoxyphenyl)methylene)bis(3-hydroxy-5,5-dimethylcyclohex-2-en-1-one) (53 mg, 0.135 mmol) and 4:1 ethanol:water mixture (2 ml) were added to a small test tube. The clear reaction mixture was heated at reflux until the starting materials dissolved, which occurred at 220 °C. The reaction mixture was removed from the heat and HCl (1 drop) was added, causing the solution to turn pink. The reaction mixture was heated at reflux for an additional 5 minutes, then cooled to room temperature. The bottom of the test tube was scratched using a glass rod causing crystals to form. The reaction mixture was cooled in an ice bath for 10 minutes, the crystals were isolated using a Hirsch funnel, washed with ice-cold 4:1 ethanol:water mixture, and dried by drawing a vacuum through the filtrate. The yield of the resulting crystal product, 9-(2-methoxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene -1,8(2H)-dione, was measured (22 mg, 42.83 %). The melting range of the product was measured to be between 189.3 and 190.7 °C.

[1]H NMR (400 MHz, $CDCl_3$) δ 0.7-1.3 (s, 3H), 1.2-1.4 (s, 4H), 1.2-1.4 (s, 4H), 1.4-1.7 (s, 1H), 2 (s, 12H), 6-8 (d, J = 6-10 Hz, 1H), 6-8 (dd, J = 6-10 Hz, 1H), 6-8 (dd, J = 6-10 Hz, 1H), 6-8 (d, J = 6-10 Hz, 1H)

III. DISCUSSION

RESULTS AND DATA ANALYSIS

The melting range of the 2-methoxybenzaldehyde reactant was measured to be between 38 to 40 °C. The mass recovered of the clear crystal dimethone derivative product from the first reaction (4) was measured to be 372.9 mg which represents a yield of 257.54 %. The melting range of this dimethone derivative was measured to be between 187 and 189.4 °C. The mass recovered of the clear crystal dimethone anhydride derivative product from the second reaction (7) was measured to be 22 mg which represents a yield of 42.83 %. The melting range of the dimethone anhydride derivative was measured to be between 189.3 and 190.7 °C.

To determine the structure of the aldehyde used in the experiment, the melting point ranges of the dimethone and the dimethone anhydride derivatives were compared to literature value. The dimethone derivative formed in the first reaction had a melting range between 187 and 189.4 °C, by comparing this to literature values of melting points ranges, the potential aldehydes used in this experiment were either 4-nitrobenzaldehyde (188-190 °C), 4-hydroxybenzaldehyde (188-190 °C), or 2-methoxybenzaldehyde (187-188 °C). The dimethone anhydride derivative formed in the second reaction had a melting range between 189.3 and 190.7 °C, by comparing this to literature value of melting points ranges, the aldehyde used in this experiment was confirmed to be 2-methoxybenzaldehyde (190-191 °C). Additionally, the melting range of the aldehyde reactant was measured to be between 38 to 40 °C, the literature value of the melting range of 2-methoxybenzaldehyde is 39 °C,[5] this falls into the range of 38 to 40 °C further verifying that the aldehyde reactant is 2-methoxybenzaldehyde.

The 257.54 % yield of dimethone derivative product was significantly too high. One reason for this high yield is due to incomplete drying of the crystals, which caused their mass to be higher than expected. To prevent this error, the crystals should be left to dry under the vacuum on the Hirsch funnel for a longer period of time, until the crystal are fully dry. Alternatively the crystals could be dried in a drying over with a temperature set below the expected melting point. Another potential cause of the high percent yield is that the reaction was incomplete and unreacted dimedone and or aldehyde could have precipitated with the product and not been removed during the wash, thus accounting for an increased mass and yield. To avoid this error, the resulting crystals should be thoroughly washed to ensure that any impurities are removed from the product. The melting range of the dimethone derivative product was slightly higher than literature value. Reasons for this include the melting point apparatus heating too quickly and causing the thermometer to show a higher temperature than the actual temperature of the heating block, the thermometer could have been calibrated incorrectly, or this could be due to human error in recording the measurement. To remove this error the thermometer should be calibrated before use, the melting point apparatus should increase temperature at a slow rate for an accurate measurement, and measurements should be taken precisely.

The percent yield of the dimethone anhydride derivative product was calculated to be 42.83 %. This yield is lower than expected, a potential reason for this is transfer loss where crystals were lost when transferring them from the test tube to the Hirsch funnel and from the Hirsch filter to the weigh boat. To improve transfer from the test tube to the Hirsch funnel, the test tube should be washed with 4:1 ethanol:water solution and then poured onto the Hirsch funnel. To improve transfer from Hirsch filter to the weigh boat the crystals should be thoroughly removed from the filter using a metal spatula. Another potential reason for the low yield is that the reaction was incomplete. To improve this the reflux should be more rigorous and for a longer period of time.

Overall, by implementing the suggested improvements of the errors listed above, the yield of both products can be improved.

IV. BIBLIOGRAPHY

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